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License Plate Recognition System

Project Report

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ABSTRACT

License Plate Recognition (LPR) is technology used to identify vehicles through their license plate. This technology applies several image processing methods and algorithms to achieve the goal. Typically LPR consist of several sequence procedures start from detection the vehicle's license plate and extracted it, and then recognize the characters of the plate using optical character recognition technique. In this project we are going to build LPR software to make it works on Saudi Arabia's license plate. All the procedures and methods should be analyzed and designed. All the techniques and algorithms should be study and choose the suitable to our software. The final software must be testing and calculate how successful is the software.

Acknowledgement

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1 Introduction

1.1 Background of Study

Nowadays License plate recognition system (LPR) becomes an important application especially for transportation. Instead of using law enforcement to monitor the increasing numbers of cars in Saudi Arabia, LPR is a more suitable in terms of security purpose for vehicles. LPR is an image processing technology that uses to identify vehicle by their license plate [1]. It can be used in many applications such as entrance admission, security, parking control, road traffic control and also border control. One of the advantages of using this system is that all cars already have their own identification by just referring to their registration plate. This means that there is no need to add transmitter or special sign to the vehicles. Optical character recognition (OCR) is the heart of LPR system [2, 3], whereas image of the car plate will go through the process of image processing before the car plate can be recognize.

1.2 Problem Statement

Due to the increasing numbers of automobiles on the road today, it is difficult to monitor the law enforcement and security for vehicles [4]. For example in 'Saher System' when the violated vehicles captured using the camera, the time needed to register the violation ticket in the system as best case it takes 24 hours because it controlled and operate by humans. Simple solution for this problem is to build a system that can automate this process so we reduce the time consuming and human can only employed to make sure that this process is running well. This problem motivate me to work on this project to simply try to build such system to identify vehicles' plates using known techniques and algorithms in image processing field.

1.3 Objective of the Project

The objectives of the project, simply as the following:

- To develop a system that can detect the location of license plate and extract it.
- To develop a system that can recognize all the characters on the license plate including the Arabic and English characters.

1.4 Scope of the Project

The following are the scopes of the project:

- The system will works on standard Saudi Arabia license plate which consist of two lines and four areas. The plates which consist of four numbers and three alphabets of Arabic and English characters.
- The project targeted to produce the LPR software. So, we will not be focusing on the hardware part like the camera for example.
- The system will process captured image. So, we will not focusing on the way the image will be taken.

1.5 Thesis Outline

This thesis consists of five chapters. In the first chapter, it discuss about the introduction, objectives and scope of the project. In Chapter 2, we will discuss more on theory and literature reviews that has been done. It well discuss about the techniques used starting from detection of the plate till recognizing the characters. In Chapter 3, the discussion will be on the implementation of the project and work that done on it and implementing the code of the LPR software. The result and discussion will be discussed in Chapter 4. Finally, in Chapter 5, conclusions and recommendations for future work are presented.

2 Literature Review and Theory

This chapter will discuss about the theory behind the project, the techniques used include the algorithm as well as the literature reviews when doing research for this project.

2.1 License Plate Recognition System Overview

LPR System consists of three major components. First, the camera that will capture and take the vehicle's image. Second, the LPR Software which will analyze and do the image processing to enhance the image and locate the plate and recognize the characters. Third, the computer that will be used to install the LPR Software on it. It could be simple PC or server that accepts request and replay the result to the requester remotely using any kind of networks. (Figure 1) shows the most common modules that used in any LPR System. (Figure 2) is example of camera that used in LPR System during acquisition image module. (Figure 3) shows example LPR Software interface.

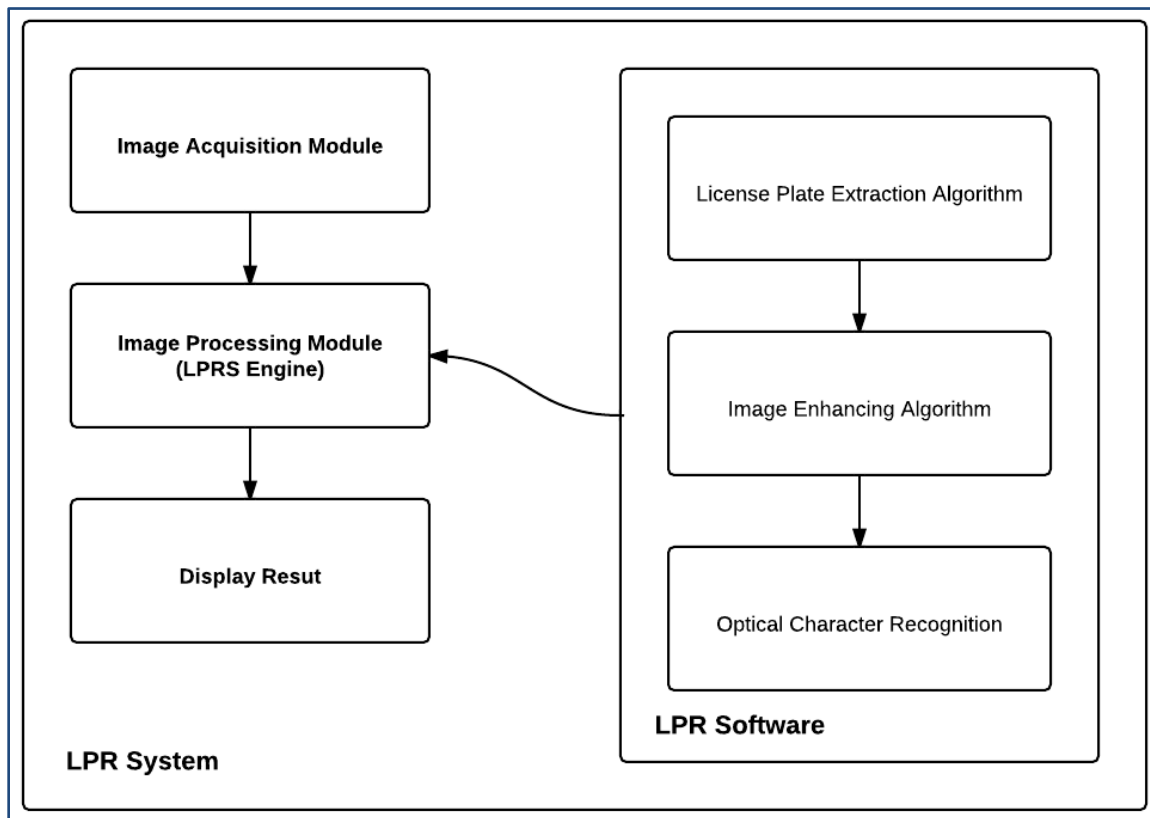


Figure 1: LPR System Block diagram

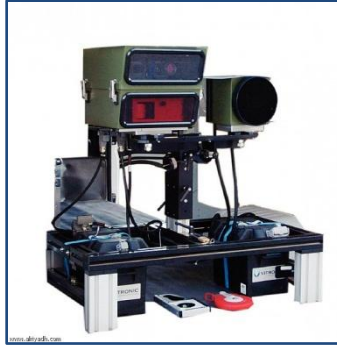


Figure 2: Mobile LPRS's Camera [5]



Figure 3: Interface of LPRS [6]

2.2 License Plate Recognition Software

Basically, LPR software consists of three phases [7]:

- 1) Detecting the location of the license plate (LP), namely, LP localization (LPL).
- 2) Segmenting the characters inside the LP, namely, character segmentation (CS).
- 3) Recognizing the meaning of the characters, namely, character recognition (CR).

According to Jing-Ming Guo and Yun-Fu Liu [7], they work on Taiwan's LP. (Figure 4) Shows the structure of Taiwan's LP and we can see that it's different from Saudi Arabia's LP (Figure 5).



Figure 4: Taiwan's LP [7]



Figure 5: Saudi Arabia's LP

In our project and regarding to the structure of Saudi Arabia's LP, we will have the same number of phases but a little different in the content of each phase. So, we will have the following phases:

- 1) License plate detection and extraction.
- 2) Normalization and segmentation of the license plate.
- 3) Segmentation and recognition of the characters.

Process of LPR begins with the getting the image this process named Image Acquisition. Image acquisition involved the technique of capturing an image from hardware like camera or video. In our project, we will assume that the image is already stored for example in our hard disc. Then, the process continues to plate detection and extraction and normalize and segment the plate. The system uses the sequence of image processing techniques such as edge detection, histogram equalization, image median filtering and also image threshold. Last but not least, each character on the car plate will be segmented, and the process of identifying the characters on the car plate is known as the optical character recognition (OCR). (Figure 6) shows an abstract view of each module that involved inside the LPR software.

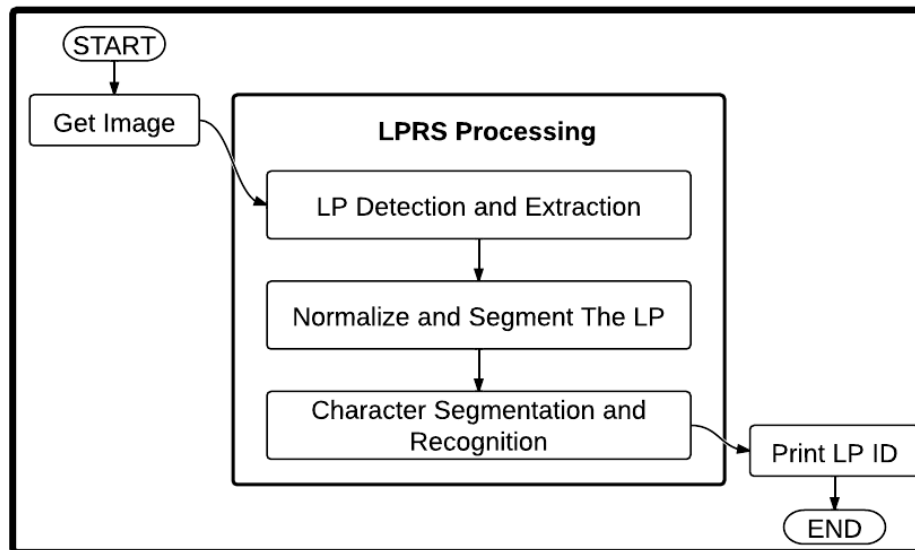


Figure 6: Abstract View of the LPR Software Flow

In the next sections we will see the algorithms used in each of the LPRS phase in details.

2.2.1 License Plate Detection and Extraction

Mainly, this phase consists of two main steps. First, detection of the plate and locate it in the image. Second, is extracting the plate from the image. Actually the locating of the plate (LPL) is the difficult part here. The main difficulty of the LPL is the environment. For example, outdoor parking spaces have at least one more problem than indoor parking spaces because of the influence of sunlight. Hence, the performance of most recognition systems varies a lot according to the energy of the sunlight. In our project, histogram equalization is adopted to reduce the sunlight or low-contrast problems. The other solution is the well-known contrast stretching, which is also capable of improving the low-dynamic-range image. However, it performs poorly in back lighted conditions, which can be totally solved with histogram equalization [7].

Process of plate detection and extraction involved the following process:

- 1) Histogram equalization.
- 2) Median filtering.
- 3) Image threshold.
- 4) Edge detection.

2.2.1.1 Histogram equalization

Histogram equalization is an image processing technique that used to contrast the image using its histogram [10]. It is also an image transformation that computes a histogram of every intensity level in a given image and stretches it to obtain a more sparse range of intensities [2]. This manipulation yields an image with higher contrast than the original. The candidate selection process utilizes the guaranteed contrast between the characters and the license plate. Therefore, it is crucial for us to contrast the image at preprocessing stage. In order to increase the contrast of the gray scale image from the PC, histogram equalization is used. The following figure shows the difference between normal image on gray scale (Figure 7.a) and after applying the histogram equalization (Figure 7.b).

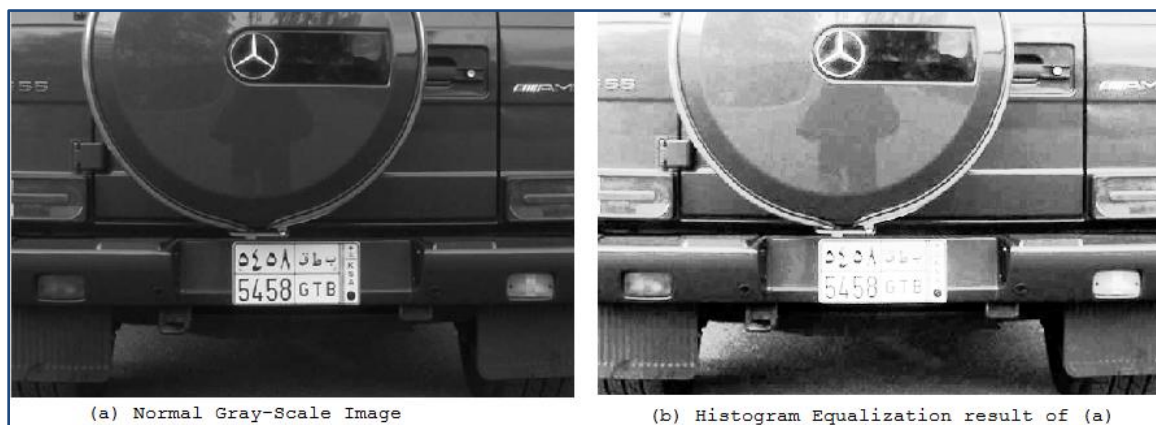


Figure 7: Effect of Applying Histogram Equalization

2.2.1.2 Median Filtering

Median filtering is a nonlinear operation often used in image processing to reduce "salt and pepper" noise [19]. Median filtering technique selects the median value from each pixel's neighborhood. Median values can be computed in expected linear time [21]. The following figure shows the different between normal image on gray scale (Figure 8.a) and after applying the median filtering (Figure 8.b).

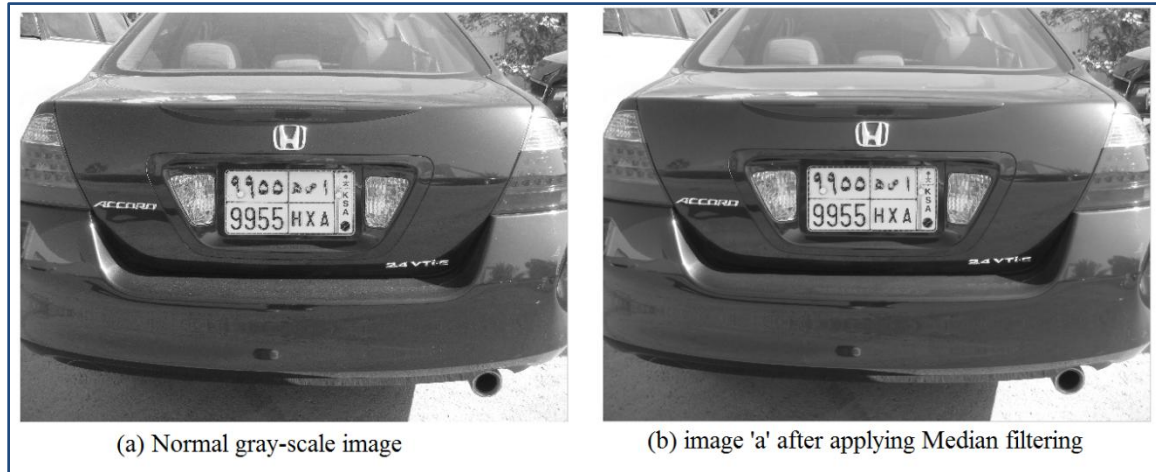


Figure 8: Effect of Applying Median Filtering

Clearing the noise using median filtering helps a lot on make the edges clear in the image. (Figure 9.a) and (Figure 9.b) are another example of using and applying median filtering to remove the noise from the image.

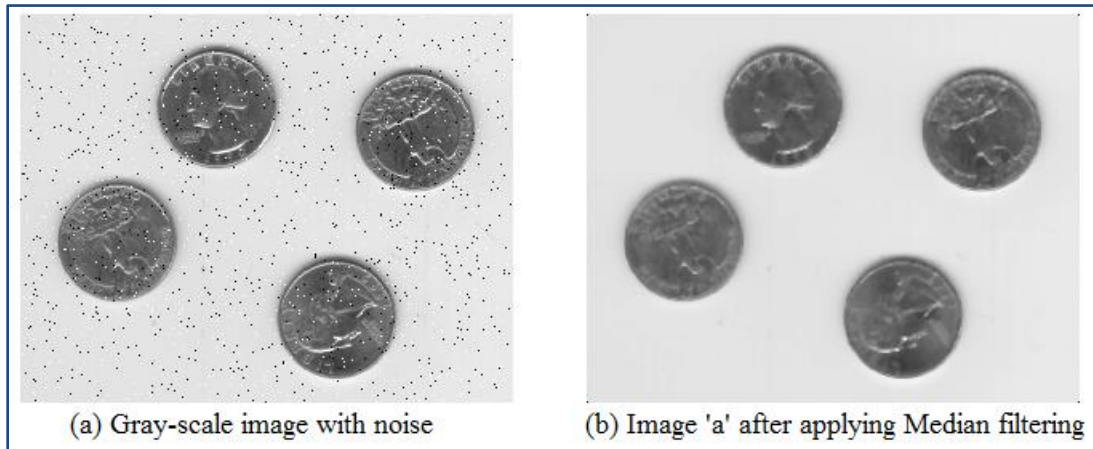


Figure 9: Example of removing noise using median filtering [19]

2.2.1.3 Image Threshold

In many applications of image processing, the gray levels of pixels belonging to the object are substantially different from the gray levels of the pixels belonging to the background. Thresholding then becomes a simple but effective tool to separate objects from the background. Examples of thresholding applications are document image analysis, where the

goal is to extract printed characters [9]. Pixels are classified into two categories when process of image thresholding occurs. Actually the result of image threshold is Binarization. Image will convert to black and white pixels. There are two types of thresholding: global thresholding and local thresholding [10]. Global thresholding also known as fixed thresholding. In fixed thresholding, the threshold value is held constant throughout the image.

$$g(x, y) = \begin{cases} 0 & f(x, y) < T \\ 1 & f(x, y) > T \end{cases}$$

Meanwhile, local thresholding depends on the position of the image. The image is divided into overlapping sections which are threshold one by one.

2.2.1.4 Edge Detection

Edge detection is an image processing technique that used to detect the edges of the picture [10]. There are several types of edge detection algorithms. Edges are the significant local changes of intensity in an image and edges typically occur on the boundary between two different regions in an image. The goal of edge detection is to identify the important feature that will be useful for plate extraction step. Actually in edge detection, there are four step involved. The followings are the step of edge detection.

- **Smoothing:** the edge detection process of eliminating as much noise as possible.
- **Enhancement:** the edge detection process of applying a filter to enhance the quality of the edges in the image (sharpening).
- **Detection:** the edge detection process of determining which edge pixels should be discarded as noise and which should be retained (usually, thresholding provides the criterion used for detection).
- **Localization:** the edge detection process determining the exact location of an edge. Edge thinning and linking are usually required in this step.

Some of the common edge detection algorithms are the Sobel, Kirsch, Laplacian , Marr (zero crossing), Roberts, Prewitt [12] and Canny operators. In LPR system, the two most popular methods for edge detection are Sobel [2, 4, 11] and Canny operator. For Saudi Arabia standard type of license plate, the most suitable algorithm should be used is Sobel. It is due to Saudi Arabia's license plate consist of a row of black characters on a white background, so we can say that the license plate region is characterized by a row of transitions from white to black and vice versa, such transitions are called "edges" [11]. As shown in (Figure 10) we can see the different result of applying different edge detection algorithms.

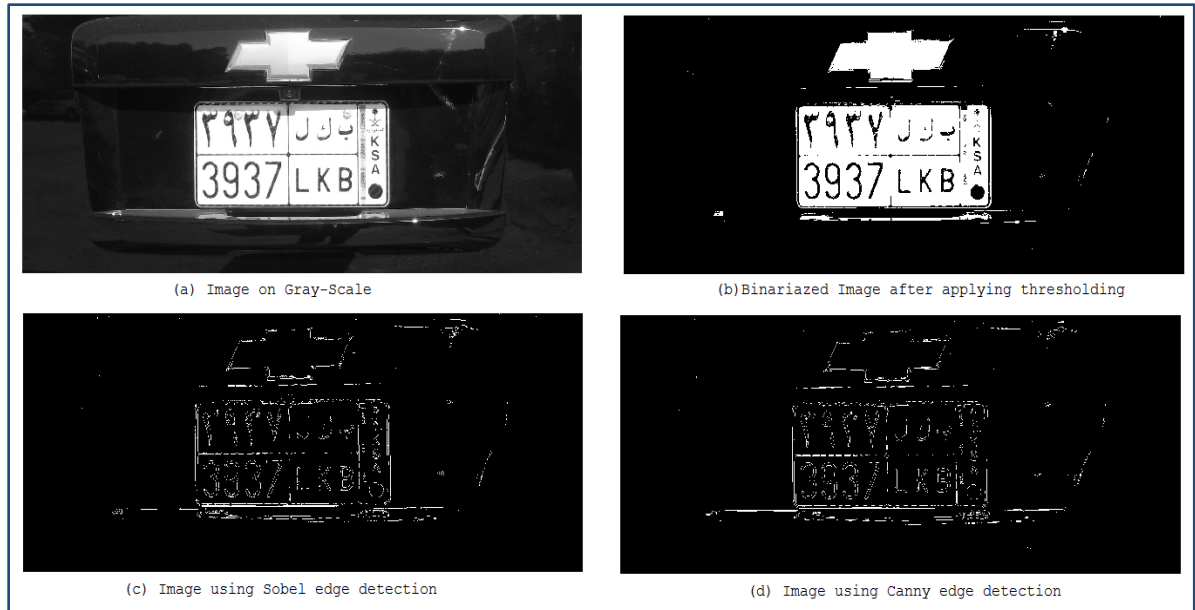


Figure 10: Result of Applying Different Edge Detection Algorithm

We apply the Sobel Detection, in which the Sobel operator performs a 2-D spatial gradient [14] measurement on an image and thus, it emphasizes regions of high spatial frequency that correspond to edges. Typically it is used to find the approximate absolute gradient magnitude at each point in an input grayscale image. The operator consists of a pair of 3x3 convolution kernels as shown in (Figure 11). The value of one kernel is simply equal to value of the other rotated by 90 degree [13].

-1	0	+1	+1	+2	+1
-2	0	+2	0	0	0
-1	0	+1	-1	-2	-1
Gx			Gy		

Figure 11: Masks used by Sobel Edge Detection [15]

2.2.2 License Plate Normalization and Segmentation

Regarding to the complexity of the special structure of Saudi Arabia's LP we have to apply process to simplify normalize this structure to make it easy for the next step which is recognition of the letters and numbers in the plate. (Figure 12) shows that Saudi Arabia's LP consist of five different areas, four that contain the characters and the identification of the LP and one that contains information about the country and the type of the LP.



Figure 12: Saudi Arabia LP Segmentation (a)

For this situation we have to segment the LP into these five areas. So, to do that there is method to segment image into specific objects using horizontal and vertical histograms projection of the binary image [16]. (Figure 13) shows example of using histograms horizontal and vertical projection on binary image of a one line license plate to segment the characters of the plate.

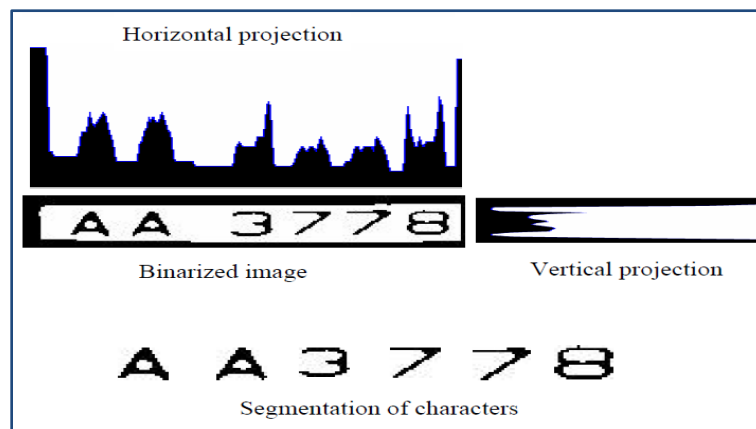


Figure 13: Using Histogram H&V Projection on Binary Image for Segmentation [16]

If we apply this process to Saudi Arabia's LP we can see the projections as show in (Figure 14). Segmentation using histogram and vertical projections try to eliminate and segment the image into different parts. So if we use the horizontal projection we can segment the plate into parts vertically by removing the white space from the image which will has the lowest value in the histogram. According to the limitation of time that I have to do and complete segmentation using this, my main objective on the project may not complete because implement this method takes time. So, I will move it and consider it as future work that should be complete but this technique still important because of the good result of it.

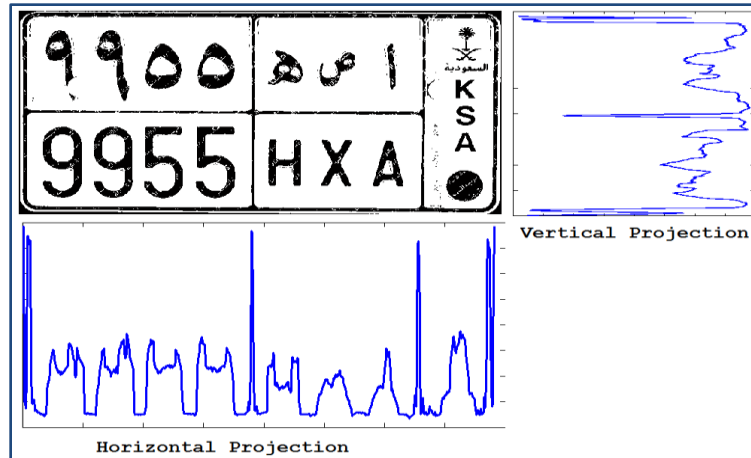


Figure 14: Applying H&V Projections on Binary Image of Saudi Arabia's LP

So, for now I will use simple clipping and cropping methods based on the percentage of each area in the LP. To apply this we have to get through three phases as shown in (Figure 15, 16, 17). In (Figure 15) we can see that the badge in the right which is not useful to us takes approximately $100 \times (123/782) = 15.7\%$ of the width of the LP from the right side and it should be eliminated. In (Figure 16) we can see that the up and bottom areas of the LP that separate the Arabic characters from the English characters each area occupied approximately $100 \times (196/392) = 50\%$ of the height of the LP. In (Figure 17) we can see that the two areas that contain the alphabet characters take approximately $100 \times (282/659) = 42.7\%$ of the width from the right of the area, and the two areas that contain the numeric characters take approximately $100 \times (377/659) = 57.2\%$ from the left. According to these parameters and information we can develop algorithm that do this job to any license plate of this structure.

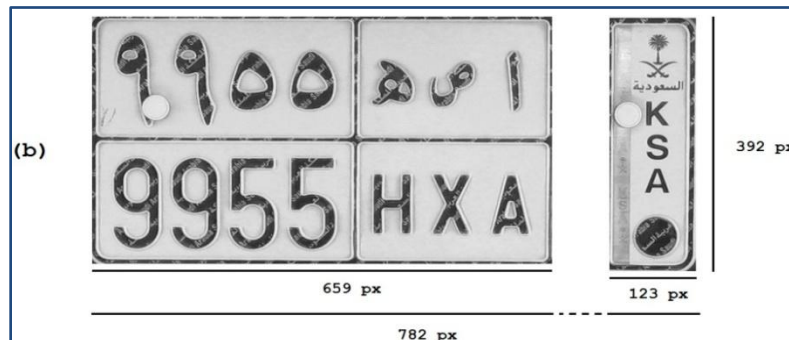


Figure 15: Saudi Arabia LP Segmentation (b)

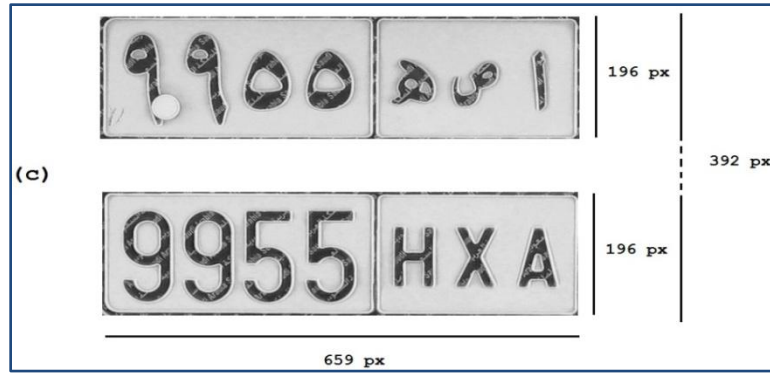


Figure 16: Saudi Arabia LP Segmentation (c)

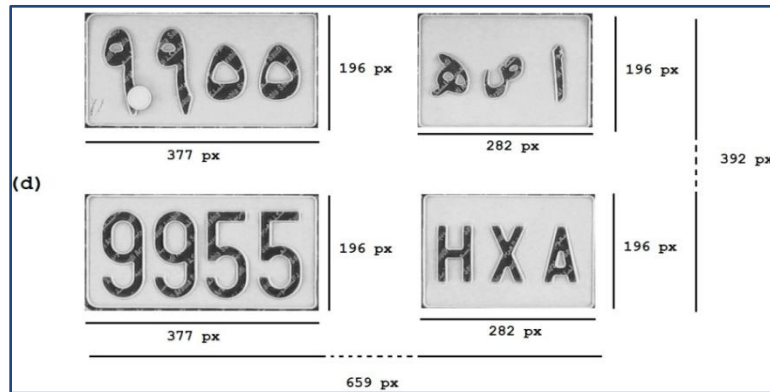


Figure 17: Saudi Arabia LP Segmentation (d)

2.2.3 Character Segmentation and Recognition

This phase contains two sub-phases. First, character segmentation, and second, character recognition.

2.2.3.1 Character Segmentation

Character segmentation is the image processing technique that used to separate or isolate each character on the car plate. The method that usually been used for character segmentation as we said in the previous section is horizontal and vertical projection [2]. The function of this operation is to isolate each character from the plate. (Figure 18) shows the horizontal and vertical projection of one area of the Saudi Arabia's LP. But for us we will be using another method that search and locate each connected objects and components. This method and algorithm called 'Labeling Algorithm' [17, 18].

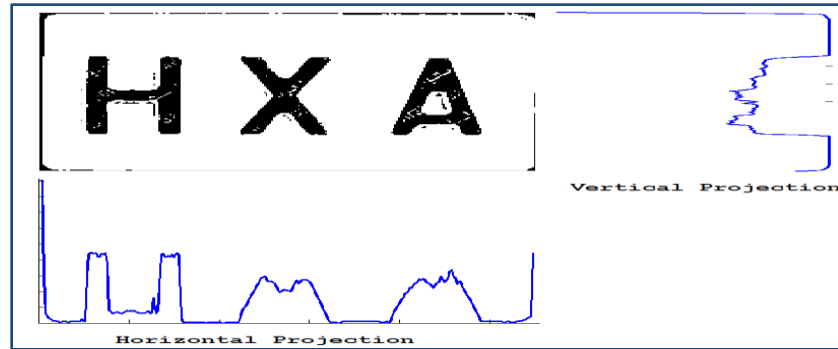


Figure 18: Horizontal & Vertical Projection

2.2.3.1.1 Labeling Algorithm

Pixel connectivity, a morphological processing starts at the peaks in the marker image and spreads throughout the rest of the image based on the connectivity of the pixels. Connectivity defines which pixels are connected to other pixels. A group of pixels that connected based on Connectivity types called an Object [17].

More about Connectivity in an Image, the following figure (Figure 19) lists all the standard two dimensional connectivities [17].

Selecting Connectivity, the type of neighborhood that may choose affects the number of objects found in an image. For example (Figure 20), if you specify a 4-connected neighborhood, this binary image contains three objects; if you specify an 8-connected neighborhood, the image has one object [17].

SUPPORTED CONNECTIVITY		
Two-Dimensional Connectivities		
4-connected	Pixels are connected if their edges touch. This means that a pair of adjoining pixels are part of the same object only if they are both on and are connected along the horizontal or vertical direction.	
8-connected	Pixels are connected if their edges or corners touch. This means that if two adjoining pixels are on, they are part of the same object, regardless of whether they are connected along the horizontal, vertical, or diagonal direction.	

Figure 19: Standard 2-D Connectivities [17]

0	0	0	0	0	0	0
0	1	1	1	1	0	0
0	1	1	0	0	0	0
1	0	0	1	1	0	0
0	0	0	1	1	0	0
0	0	0	0	0	0	0

A binary image

Figure 20: Binary Image Example [17]

2.2.3.2 Character Recognition

There are few methods that can be used for character recognition. The following are among the most common [22]:

- 1) Neural Network based method.
- 2) Template Matching.

2.2.3.2.1 Neural Network based method

Neural network is a mathematical model for information processing base on connectionist approach to computation [4]. Neural network is also one of the common and popular approaches in application like pattern recognition, times series prediction, function approximation, clustering, etc. Neural network is a good method and give better result but in license plates, the characters are take the same shape same font so no need to use neural network method because of its complexity and the time needed to learn the system.

2.2.3.2.2 Template Matching

Template matching is basically the technique that evaluates the similarity between the image and a set of reference image called template. On the other words, template matching is the techniques compare portions of images against one another. The template with the highest matched point will be the character appears in the image [2]. If standard deviation of the template image compared to the source image is small enough, template matching may be used. Templates are most often used to identify printed characters, numbers, and other small, simple objects. This is why it is the most suitable algorithm to use in our project.

The matching process is done by moving the template image to all possible positions in a larger source image which is the template and compute a numerical index that indicate how well the template matches the image in that position. Furthermore, matching is done on a pixel-by-pixel basis.

Template matching has two major problems to tackle. First the template matching is sensitive to image distortion. Distortion can be occurring during image acquisition process [2]. For example, the distortion is due to variations in viewing angle, diffused light, screws, and lightning and also bolt that distort the view, and a limited range in view. One way to

reduce distortion during template matching is by making a lot of samples for one extracted binary character image.

Finally, to recognize an image, it needs to be correlated with all templates in the reference set. Correlation is a measure of the degree to which two variables agree, not necessary in actual value but in general behavior. The two variables are the corresponding pixel values in two images, template and source. (Figure 21) shows example of template used in template matching algorithm.

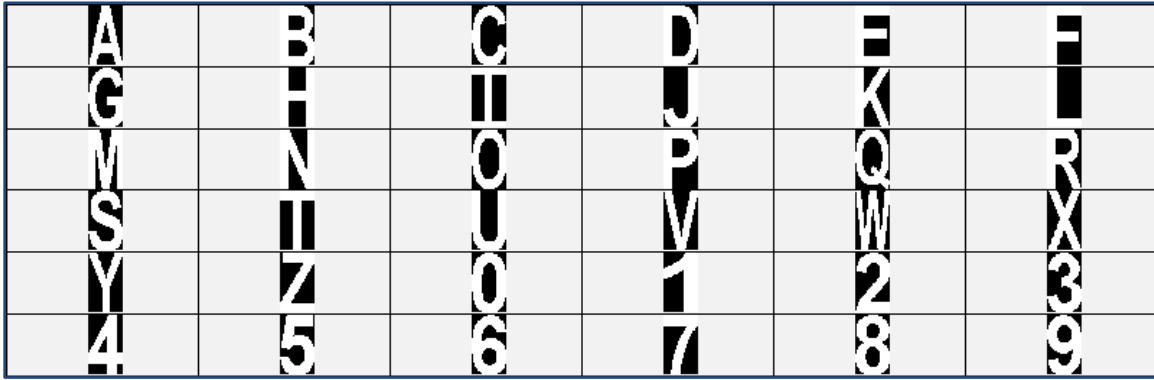


Figure 21: Examples of English Letters and Numbers Template

According to [19], correlation is closely to convolution, whereby convolution is a neighborhood operation in which output pixel is the weighted sum of neighboring input pixels. Different with convolution, in correlation, the value of an output pixel is computed as a weighted sum of neighboring pixels except that the matrix of weights, in this case called the correlation kernel, is not rotated during the computation. Correlation can be used to locate features within an image; in this context correlation is often called template matching.

$$r = \frac{\sum_m \sum_n (A_{mn} - \bar{A})(B_{mn} - \bar{B})}{\sqrt{\left(\sum_m \sum_n (A_{mn} - \bar{A})^2 \right) \left(\sum_m \sum_n (B_{mn} - \bar{B})^2 \right)}}$$

where $\bar{A} = \text{mean2}(A)$, and $\bar{B} = \text{mean2}(B)$.

Figure 22: 2-D Correlation Coefficient formula [19]

2.3 Real Life Applications

There are many applications of the car plate recognition system. This system is applied in access control system, border control and car park system. In a car park situation, the system compares the recognized license plate number with is a list of authorized users or people with a membership and allows entry if a match is found. This information will be contained in a database. Alternatively, the recognized number is stored with the time of entry. When the user exits, the number is read again and the parking fee is calculated from both entry and exit times. Long term parking at the airport is also a good candidate for the use of an LPR system as the actual time of entry can retrieve in the event of a lost ticket that was issued. In border control situation, the LPR system can be used to double check that the car is allowed to cross and shorten the time it takes to go through the border, hence shortening the typical long queues. Each vehicle would be registered in a central database with information about the car and the drivers. Moreover, all border crossings can be recorded automatically and used for statistics or law enforcement. (Figure 23) show some example of the applications of LPR system.

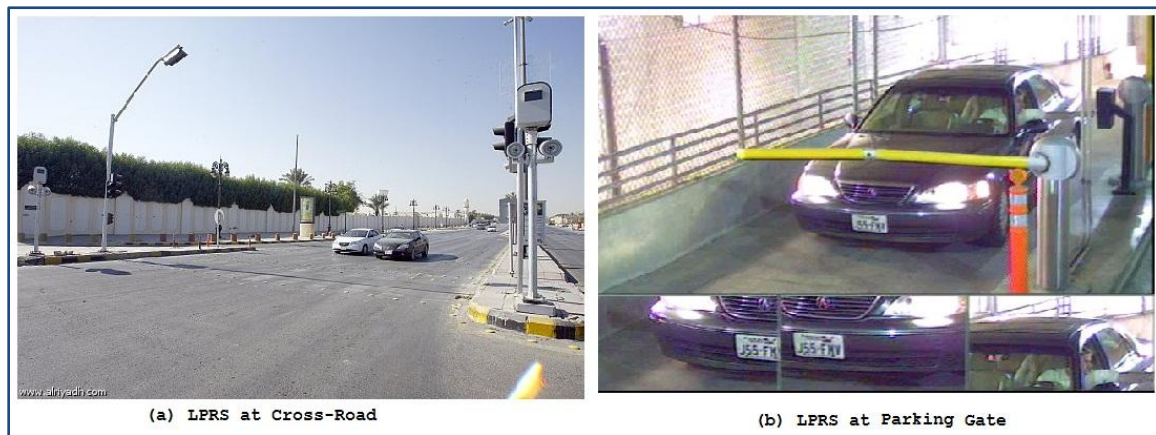


Figure 23: Real Application Example of LPRS

3 Implementation

In this chapter I will try to show and explain the works that I have done in the project. In the first section I will show the work plan that I follow till finishing the project. In the second section, I will talk about MatLab which is the powerful tool that I used to study the LPR system and produce the software. And in the final section, I will try to show and explain each components and procedures that have done in the LPR software.

3.1 Work Plan

Working on this project can be divided into seven activities, from the star till the end of it. The following table (Figure 24) shows these activities and start-date and end-date of each activity. We can see that working in this project takes almost 4 months. (Figure 25) shows project's activities represented as Gantt-Chart using OpenProj application.

	Activity Name	Start Date	End Date	Duration
1	Read Papers	21-Sep-11	15-Oct-11	18 Days
2	Read Papers	11-Nov-11	20-Dec-11	28 Days
3	Learn MATLAB	8-Oct-11	16-Nov-11	28 Days
4	Analyze The Problem	18-Nov-11	23-Nov-11	4 Days
5	Design The System	26-Nov-11	30-Nov-11	3 Days
6	Implementation & Test	10-Dec-11	5-Jan-11	19 Days
8	Documentation	24-Dec-11	6-Jan-11	10 Days
9	Total	21-Sep-11	6-Jan-11	110 Days

Figure 24: Work Plan Activities

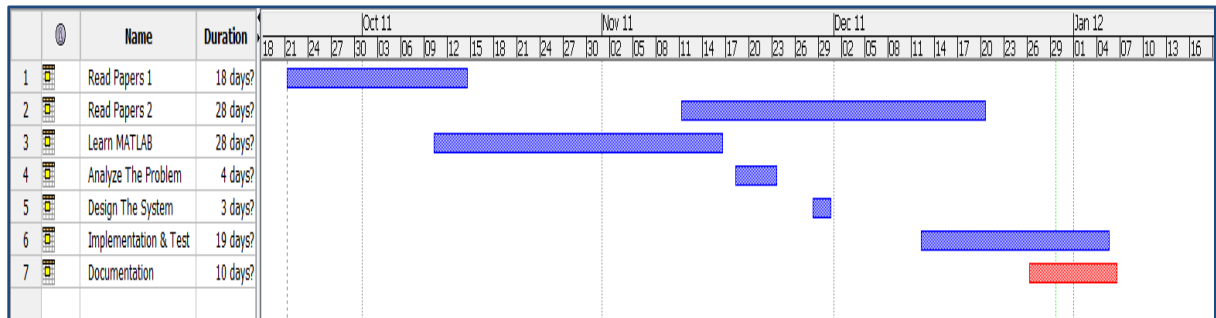


Figure 25: Snapshot of Project Gantt-Chart

3.2 MATLAB as Programming Language

Before starting the works on this project I wonder how I could do this project using the programming languages that I know C/C++ and Java. But I was directed [23] to do it using Matlab and I have to learn it first. And after reading some resources I found that Matlab is the most suitable for my project. Because I can concentrate on the algorithms and manipulate them quickly and the main idea of the project is to study different type of algorithms and apply it and study the result and not just implement the software.

MATLAB stands for "Matrix Laboratory". This software is developed by The MathWorks. MATLAB allows matrix manipulations, plotting of functions and data, implementation of algorithm, creation of user interface and interfacing with programs written in other languages, including C, C++, and FORTRAN. According to [20], MATLAB is a very powerful, high-level technical computing language used by mathematicians, scientists and engineers to solve problems in a wide range of application areas.

It also comes with several toolboxes to solve most common problems. There are many applications toolbox that included in MATLAB, for example, the Image Processing Toolbox software. This toolbox is a collection of functions that extended the capability of the MATLAB numeric computing environment [19]. For image processing purpose, the toolbox supports various operations, including the following:

- Image analysis and enhancement.
- Image registration.
- Neighborhood and block operations.
- Linear filtering and filter design.
- Region of interest operations.
- Spatial image transformations.
- Transforms.
- Deblurring.
- Morphological operations.

3.3 License Plate Recognition Software Components

The license plate recognition software consists of three main functions which are License Plate Detection and Extraction, Normalize and Segment the License Plate, and Character Segmentation and Recognition. So, the software will contain three main procedures and of course the main procedure which will do the calls and manage all the processes from the beginning till the end. To make the idea clear and before I write the details of each procedure (Figure 26) shows the flow of the whole system. And we can see that there two more procedures; Enhance image Using histogram equalization, and get image, these two procedures are not in the core works of the software but they still components in the software. Moreover, there is the GUI of the software.

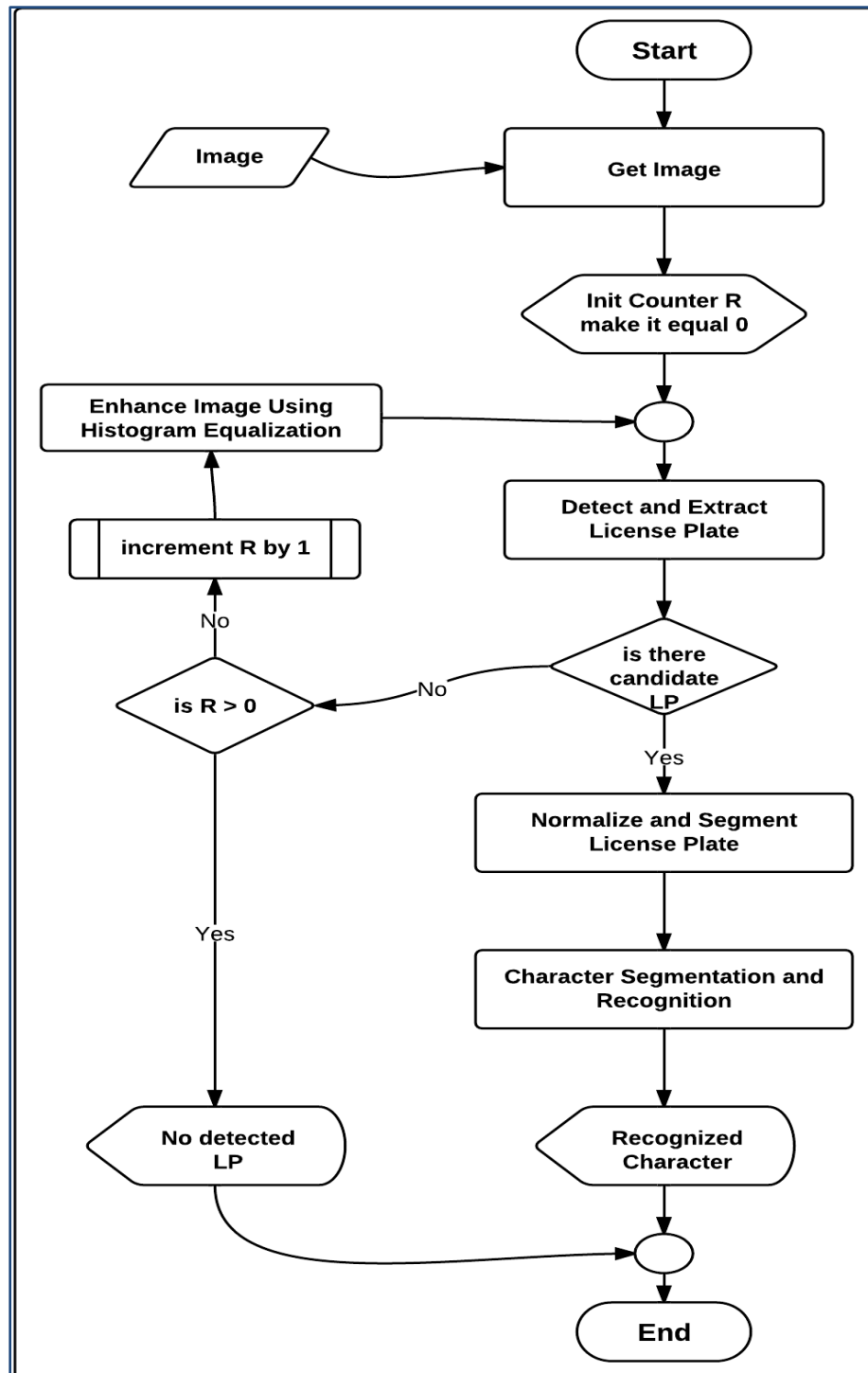


Figure 26: LPR Software Flowchart

I will clarify the different components using pseudo code. Starting the software, the job will be handled by the Main procedure.

3.3.1 Main Procedure

- Step 1: call 'Get Image' procedure to get input image from the user.
- Step 2: create counter variable 'R' and initialize it to zero.
- Step 3: call 'Detect and Extraction License Plate' procedure and pass the image to it. The return from this method should be image object that represent the candidate license plate.
- Step 4: check if 'Detect and Extraction License Plate' procedure returns object or not. If yes, Go-to step 5. If No, check if variable R is greater than zero or not. If yes, this mean we are on second round and there is no license plate detected and we have to end the process. If no, increment R by one and then call 'Enhance Image Using Histogram Equalization' procedure and pass the image to it, this method should return the same image but enhanced. Then, Go-to step 3.
- Step 5: call 'Normalize and Segment the License Plate' procedure and pass the image object that contains the candidate license plate. The return from this method should be four image objects each one represent one area of the license plates' areas.
- Step 6: call 'Character Segmentation and Recognition' procedure and pass the four image objects to it. The return from this method should be strings that represent the recognized characters.
- Step 7: print the recognized characters to the user.

3.3.2 Detection and Extraction the License Plate Procedure

- Step 1: Convert the image to gray scale image.
- Step 2: Apply median filtering using 5x5 matrix for the blocks
- Step 3: Threshold the image and apply Sobel edge detection to the binary image.
- Step 4: Remove some noise in the image by removing small objects.
- Step 5: Since the plate will be somewhere in the center of the image, we may remove the details that connected to the boarder of the image.
- Step 6: Dilate the lines that on 90 and 0 degrees on the image so it will be clear to detect the plate.
- Step 7: Fill interior gaps so most of the plate's pixels will be connected together.
- Step 8: Label connected components.
- Step 9: Measure properties of image regions that take bounding box shapes.
- Step 10: Extract all the objects that have been found.
- Step 11: Eliminate objects that don't meet with the aspect ratio of Saudi Arabia's plate which is around $0.66 > \text{aspect ratio} > 0.43$, and return the rest objects.

3.3.3 Normalization and Segmentation the License Plate Procedure

- Step 1: Get the size (width and height) of the image.
- Step 2: Crop 15.7% of the width from the image from the right side and keep the result image.
- Step 3: Get the size (width and height) of the result image from step 2.
- Step 5: Crop the image by 50% of the height and keep the result images bottom and up.
- Step 6: Get the size (width and height) of the result images from step 5.
- Step 7: Crop each image from step 5 by 42.7% of the width from the right side, and return the four results objects.

3.3.4 Character Segmentation and Recognition Procedure

Before we show the pseudo code of this procedure we have to mention the pre-processes behind it. Character recognition uses template matching method so we have to make and build these templates. We build four different templates Arabic letters, Arabic digits, English letters, and English digits. The objective of separating the characters is to achieve the most accurate recognition. In appendix A, we can see tables that corresponding to what have been used in the software. In Arabic digits, English letters, and English digits we use MS paint program to produce the characters as Bit-Map image and then we resized it to 42x24 to make the template of each one, but in Arabic letters we used real data from real image. We cleared it and extracted from the images and we made the Bit-Map of each letter. For this operation we used about 48 different license plates from different images. We have to mention that on Saudi Arabia's license plate there are only 17 letters used and 10 digits from (0-9). The following (Figure 27) shows these letters in both languages.

Letters		Digits	
Arabic	English	Arabic	English
ا	A	٠	0
ب	B	١	1
ج	J	٢	2
د	D	٣	3
ر	R	٤	4
س	S	٥	5
ص	X	٦	6
ط	T	٧	7
ع	E	٨	8
ق	G	٩	9
ك	K		
ل	L		
م	Z		
ن	N		
هـ	H		
و	U		
ى	V		

Figure 27: Characters used in Saudi Arabia's LP

This procedure will get four image objects as input and it will do the following steps to each one of them.

- Step 1: Clear the objects that connected to the boarder to make sure that only the characters will be in the image.
- Step 2: Label connected components.
- Step 3: Extract all the objects that have been found.
- Step 4: Resize the extracted objects to meet with the template's images size which is 42x24.
- Step 5: Read the template objects.
- Step 6: Compute the correlation coefficient between each extracted objects from the input image and the objects in the template.
- Step 7: Search for the template's object that has maximum correlation coefficient, and use the index of it to determine the targeted recognized character.
- Step 8: Do step 6 and 7 for each extracted objects from the input and keep the result in string objects and return it.

After getting the recognition of the characters we have to mention that Matlab doesn't support Arabic characters so I use three English characters for each one Arabic character. (Figure 28) shows these substitutions.

Letters		Letters	
Arabic Letter	English Substitution	Arabic Letter	English Substitution
ا	ALF	ق	GAF
ب	BAA	ك	KAF
ح	HAA	ل	LAM
د	DAL	م	MEM
ر	RAA	ن	NON
س	SEN	هـ	HAH
ص	SAD	و	WOW
ط	TAA	ى	YAA
ع	AEN		

Figure 28: Arabic letters displaying substitutions

3.3.5 LPR Software's GUI

For the graphical user interface we used GUIDE toolbox in Matlab R2010 to design and build the interface. The interface consist of five fields; the input image information, segmented license plate after processing, recognition texts after characters recognition phase, information about processing times, and finally push buttons to browse and choose image and to execute the processing. (Figure 29) shows the interface of the application and (Figure 30) shows the interface after select image and executes the process.

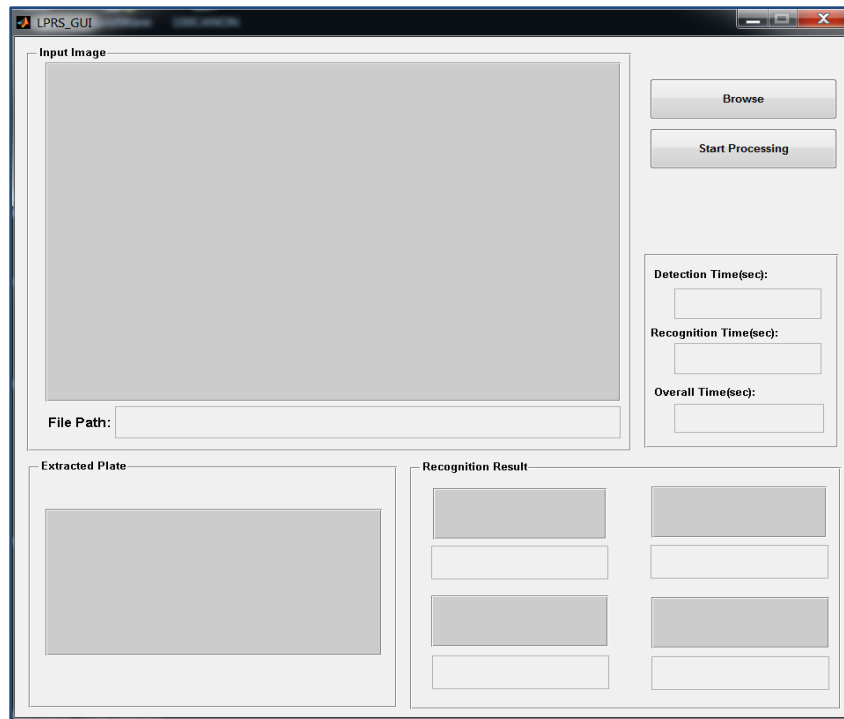


Figure 29: LPRS GUI 1

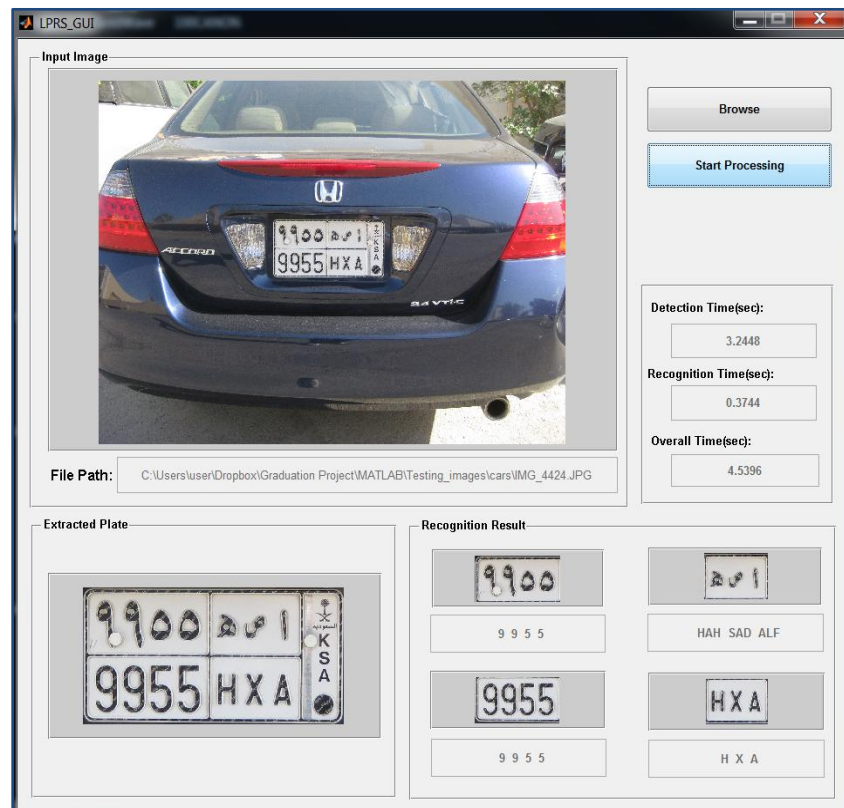


Figure 30: LPRS GUI 2

4 Testing and Result's Discussion

In this section I will try to do some tests and show the results of each test. To measure the work good we have to do different type of tests depend on the function and procedure that we want to test and measure it. For that, we will have six tests as following:

4.1 LPRS Functionality Tests

- **Detection Test:** this test applied to measure how successful the detection and extraction the license plate from the picture.
- **Character Recognition (OCR):** this test applied to get and measure the success rate of the recognition part. Since our software use different template on the template matching method each one depend on the character type we want to recognize, so we will apply this test in that way.
- **Overall System Test:** this test applied to measure the system work passing all the three major functions together which are detection and extraction the license plate, normalization and segmentation the license plate, character segmentation and recognition.

The following table shows the result after applying the tree tests:

	Test Type		No. of Samples	Success	Failure	Successful Rate	Running Time Rate (sec.)
1	Detection Test		64	52	12	81.25%	3.16
2	Character Recognition (OCR)	Arabic Letter	42	31	11	73.8%	0.27
		Arabic Digit	49	23	26	46.9%	
		English Letter	45	37	8	82.2%	
		English Digit	43	36	7	83.7%	
		OCR Test	179	127	52	70.94%	
3	Overall System Test		64	42	22	65.62%	5.05

From the three tests we can see that the first test which is detection and extraction of the plate gives the best successful rate which is 81.25%. OCR test gives 70.94% successful rate. Overall system test gives 65.62% successful rate. If we look at the four different tests for the OCR we can see that their results are in general not in the same level especially on the Arabic characters. The question here is why this happened? And the answer for that, because the Arabic letters template made of real data from real license plates, but for the Arabic digit template it made of simple printed font using MS-Paint program. So, the successful rate of OCR Arabic letter is better than OCR Arabic digit.

4.2 Real Example Tests and Discussions

Now in this section we will try to take some real examples and discuss some issues on it.

4.2.1 Example 1: Success Example

Input Image



File Path: C:\Users\user\Dropbox\Graduation Project\MATLAB\Testing_images\cars\IMG_4424.JPG

Browse

Start Processing

Detection Time(sec): 3.1512

Recognition Time(sec): 0.5616

Overall Time(sec): 4.4772

Extracted Plate



Recognition Result


9 9 5 5


9 9 5 5


H A H S A D A L F


H X A

Figure 31: LPRS Test example 1

From the result of the application we can see that the application detect and extract the plate correctly, segment the plate done correctly too, recognition and segmentation of the characters also done correctly.

4.2.2 Example 2: Failure Example

Input Image



File Path: C:\Users\user\Dropbox\Graduation Project\MATLAB\Testing_images\cars\IMG_4425.JPG

Browse

Start Processing

Detection Time(sec): 3.0888
Recognition Time(sec): 0.2184
Overall Time(sec): 4.0716

Extracted Plate



Recognition Result


3 ? 9 3 7


KAF BAA LAM


3 7 ? 9 8 ? ? ? ? ?


L ? K ? J A B A

Figure 32: LPRS Test Example 2

From the result we can see that the detection and extraction and segment of the plate done correctly but the OCR give wrong results. The problem happened because of the high brightness that focused in the plate so during the binarization of the image the characters lose their features and the shapes that identify them. So here we must minimize the enhancement of the image so we keep the characters in good shapes.

4.2.3 Example 3: Failure Example

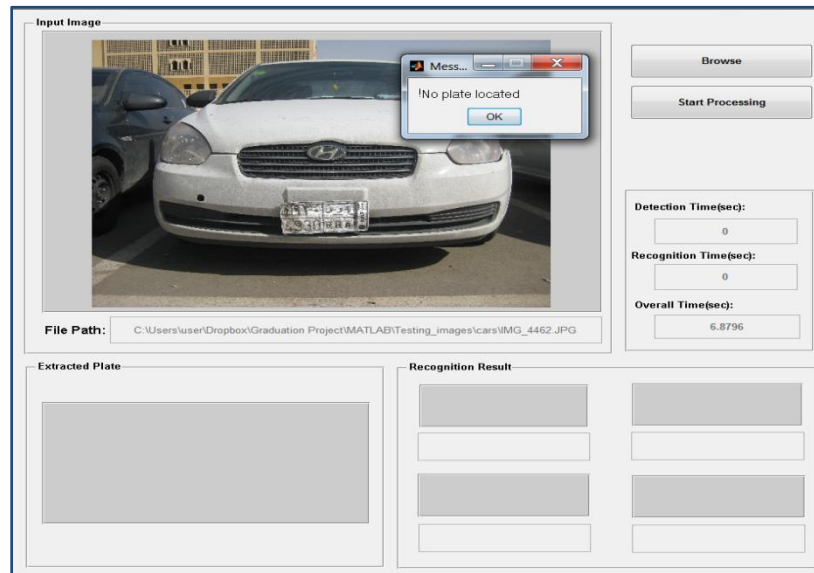


Figure 33: LPRS Test Example 3

We can see that the application fail to detect the plate and this may happen for many reasons in this situation. Because the bad situation of the license plate and the color of the vehicle which is the same color of the background of the LP the application fail to detect the edges of the plate.

4.2.4 Example 4: Failure Example

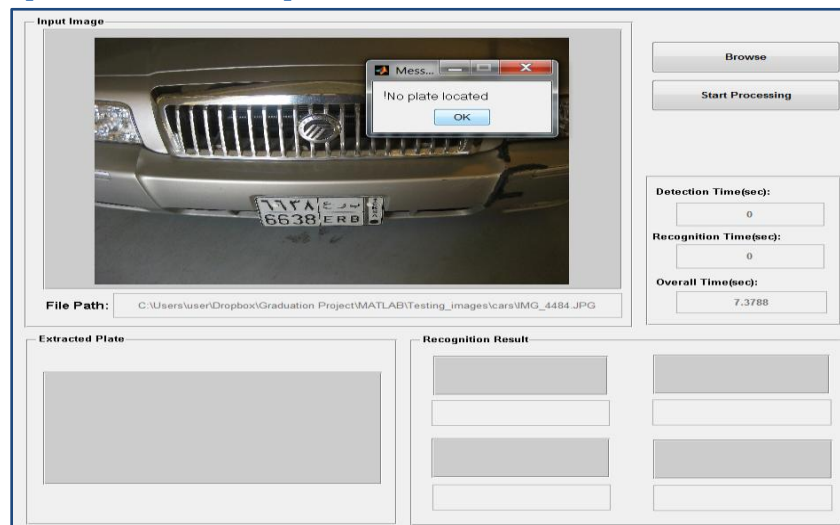


Figure 34: LPRS Test Example 4

In this example also the application fails to detect the plate because of the angel that the image was taken.

4.2.5 Example 5: Failure Example

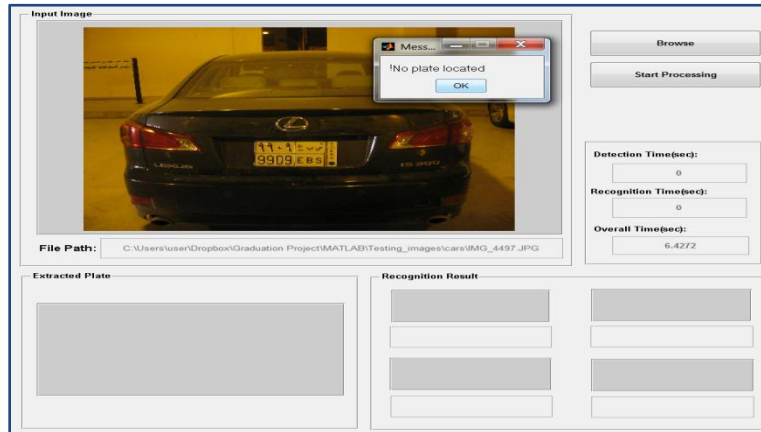


Figure 35: LPRS Test Example 5

In this example the application fails to detect because of the environment color of the image and darkness of the vehicle. As a solution we may add filtering method that process on the brightness of the image and make it brighter.

4.2.6 Example 6: Failure Example

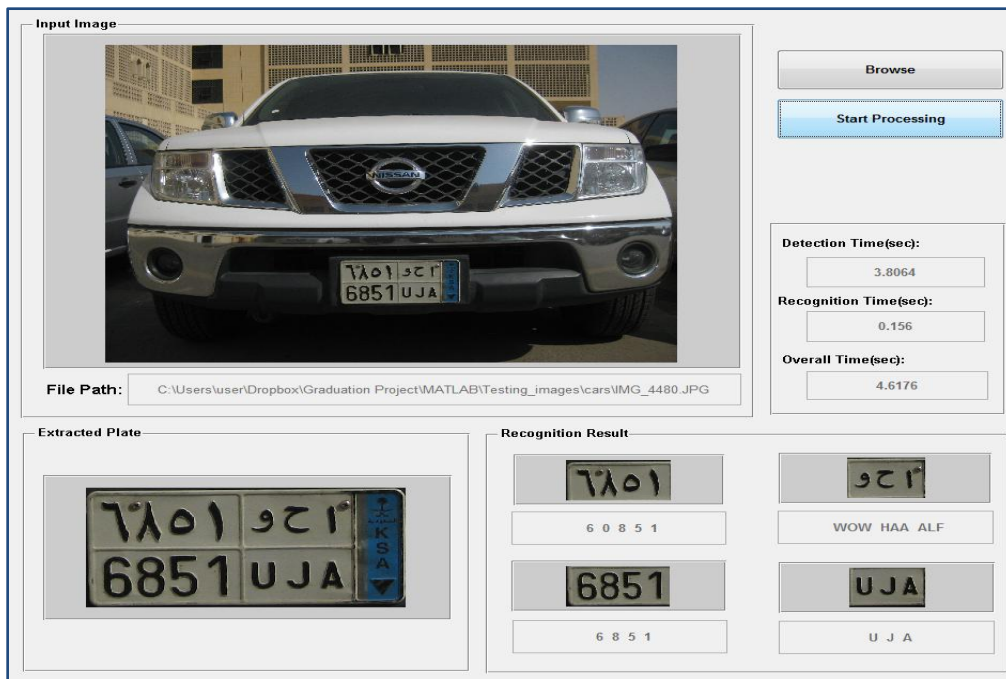


Figure 36: LPRS Test Example 6

In this example everything seems to be correct the failure happen on recognition of the Arabic digit. We can see there is insertion character in the numbers. And that's happen because in segmenting the characters from the area the application assume the screw of the plate as object within the characters. As a suggestion solution is to use different segmentation method like horizontal and vertical histogram projection.

4.2.7 Example 7: Failure Example

Input Image


Browse

Start Processing

Detection Time(sec):
3.354

Recognition Time(sec):
0.1404

Overall Time(sec):
4.1028

File Path: C:\Users\user\Dropbox\Graduation Project\MATLAB\Testing_images\cars\IMG_4499.JPG

Extracted Plate


Recognition Result

 0 0	 NON BAA ALF ALF
 4 6 0 0	 B A A

Figure 37: LPRS Test Example 7

From the results we can see that detection the plate and segment it is correctly done, but recognition the Arabic digits missing two characters which are the 4 and 6, also there is insertion of one letter in the beginning of the Arabic letters. For the Arabic digits we can see that the two first digits are connected with the shadow that appears in the above side of the plate. So during the smoothing the image they removed as they connected with the borders.

5 Conclusion and Future Work

5.1 Conclusion

As a conclusion, we can see that the software did the detection part with 81.25% successful rate which is in general good. Also the recognition part gives 70.94% successful rate. And the overall system's successful rate is 65.62%. So, we have to do more works on our application to get higher rates that insure that the application works well. The LPR systems use a lot of image processing algorithms and concepts, so it's very important that we should see and study all the options and apply the most suitable for us in our project. In this software we saw that segmentation and extraction involved in almost all of the software's phases. So, more study needed in this part so we may get better results. In general LPR system is good and useable as a real application in our life, but all aspects and cases should be taken to make sure that the system always gives the correct and most accurate results.

5.2 Future Work

The segmentation and extraction process is almost applied on all of the software process. So for that, we need to use better solution. And one of the solutions I mentioned in the report is using the horizontal and vertical histogram projection to segment the unneeded blocks from the image and the plate. Other important technique that it should be use is using the histogram identification to recognize the characters on the plate. More important thing is, If we want to use such system in real time application, implement the code using MATLAB is not good because it lacks on the performance and time, and size of we want such software to be on embedded systems. And the correct choice is to use more native and high performance programming languages like C/C++, Java, or Assembly programming language.

6 References

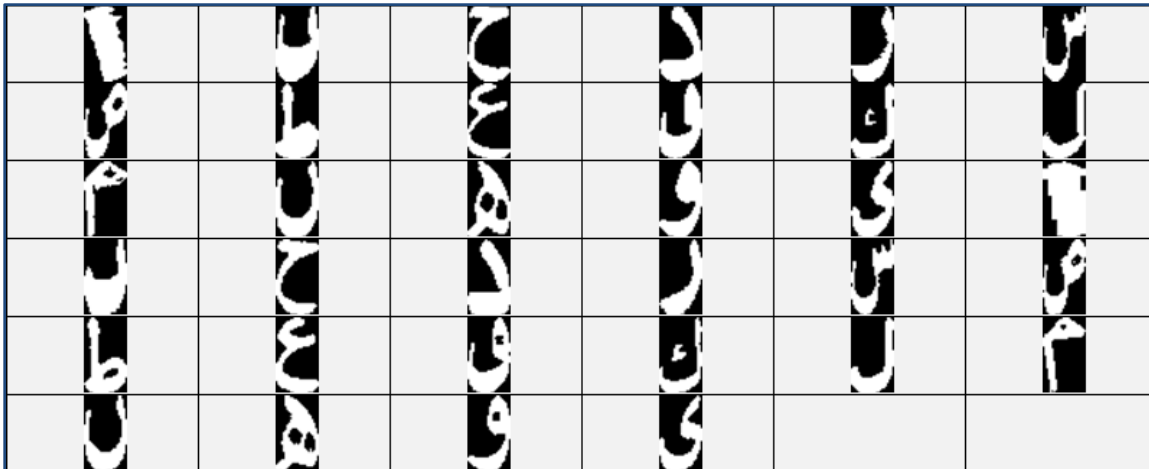
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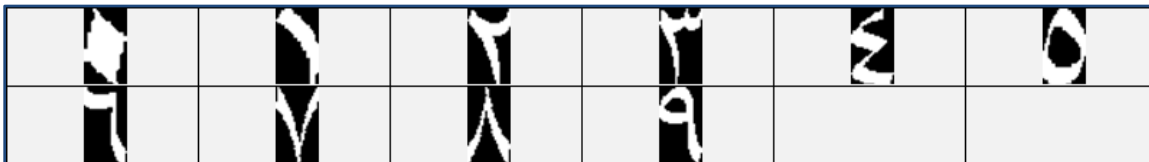
7 Appendix

7.1 Appendix A: Templates Used in Recognition

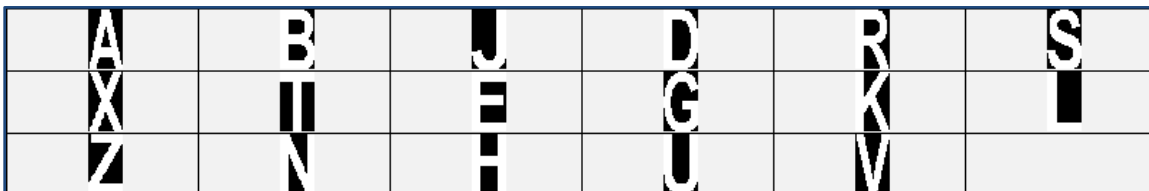
7.1.1 Arabic Letters Template



7.1.2 Arabic Digits Template



7.1.3 English Letters Template



7.1.4 English Digits Template



7.2 Appendix B: Source Code

7.2.1 Main Procedure Code (file: LPRS_GUI.m)

```
function varargout = LPRS_GUI(varargin)
gui_Singleton = 1;
gui_State = struct('gui_Name',       mfilename, ...
                  'gui_Singleton',   gui_Singleton, ...
                  'gui_OpeningFcn',   @LPRS_GUI_OpeningFcn, ...
                  'gui_OutputFcn',    @LPRS_GUI_OutputFcn, ...
                  'gui_LayoutFcn',    [] , ...
                  'gui_Callback',     []);
if nargin && ischar(varargin{1})
    gui_State.gui_Callback = str2func(varargin{1});
end

if nargout
    [varargout{1:nargout}] = gui_mainfcn(gui_State, varargin{:});
else
    gui_mainfcn(gui_State, varargin{:});
end
function LPRS_GUI_OpeningFcn(hObject, eventdata, handles, varargin)
handles.output = hObject;
guidata(hObject, handles);
function varargout = LPRS_GUI_OutputFcn(hObject, eventdata, handles)
varargout{1} = handles.output;
function browseBtn_Callback(hObject, eventdata, handles)
[filename, pathname] = uigetfile('*.JPG','*.JPEG', 'Select IMAGE');
imagePath = fullfile(pathname, filename);
set(handles.edit1,'string',imagePath);
originalImage = imread(imagePath);
axes(handles.axes1);
imshow(originalImage);
function processBtn_Callback(hObject, eventdata, handles)
clc;
overAllStart = cputime;
imagePath = get(handles.edit1,'string');
originalImage = imread(imagePath);
detectionTime = 0;
recognitionTime = 0;
overallTime = 0;
passLocalizationPhase = 0;
startTime = cputime;
candidateLP = DE_LP(originalImage);
if(candidateLP~=0)
    grayScaleImg = rgb2gray(originalImage);
    grayScaleImg = histeq(grayScaleImg);
    candidateLP = DE_LP(grayScaleImg);
    if(candidateLP~=0)
        passLocalizationPhase = 1;
        endTime = cputime;
        detectionTime = endTime - startTime;
    end
else
    passLocalizationPhase = 1;
    endTime = cputime;
    detectionTime = endTime - startTime;
end
if(passLocalizationPhase == 1)
    axes(handles.axes2);
    imshow(candidateLP);
    [LPQ1,LPQ2,LPQ3,LPQ4]=NS_LP(candidateLP);
    axes(handles.axes3);
    imshow(LPQ1);
    axes(handles.axes4);
    imshow(LPQ2);
    axes(handles.axes5);
    imshow(LPQ3);
    axes(handles.axes6);
    imshow(LPQ4);
```

```

        startTime = cputime;
        [Q1Rec,Q2Rec,Q3Rec,Q4Rec] = OCR(LPQ1,LPQ2,LPQ3,LPQ4);
        endTime = cputime;
        recognitionTime = endTime - startTime;
        set(handles.edit2,'string',Q1Rec);
        set(handles.edit3,'string',Q2Rec);
        set(handles.edit4,'string',Q3Rec);
        set(handles.edit5,'string',Q4Rec);
        msgbox('Process Done','Message');
    else
        msgbox('!No plate located','Message')
    end
    overAllEnd = cputime;
    overAllTime = overAllEnd - overAllStart;
    set(handles.edit6,'string',num2str(detectionTime));
    set(handles.edit7,'string',num2str(recognitionTime));
    set(handles.edit8,'string',num2str(overAllTime));
    function edit1_Callback(hObject, eventdata, handles)
    function edit1_CreateFcn(hObject, eventdata, handles)
    if ispc && isequal(get(hObject,'BackgroundColor'),
    get(0,'defaultUicontrolBackgroundColor'))
        set(hObject,'BackgroundColor','white');
    end
    function edit2_Callback(hObject, eventdata, handles)
    function edit2_CreateFcn(hObject, eventdata, handles)
    if ispc && isequal(get(hObject,'BackgroundColor'),
    get(0,'defaultUicontrolBackgroundColor'))
        set(hObject,'BackgroundColor','white');
    end
    function edit3_Callback(hObject, eventdata, handles)
    function edit3_CreateFcn(hObject, eventdata, handles)
    if ispc && isequal(get(hObject,'BackgroundColor'),
    get(0,'defaultUicontrolBackgroundColor'))
        set(hObject,'BackgroundColor','white');
    end
    function edit4_Callback(hObject, eventdata, handles)
    function edit4_CreateFcn(hObject, eventdata, handles)
    if ispc && isequal(get(hObject,'BackgroundColor'),
    get(0,'defaultUicontrolBackgroundColor'))
        set(hObject,'BackgroundColor','white');
    end
    function edit5_Callback(hObject, eventdata, handles)
    function edit5_CreateFcn(hObject, eventdata, handles)
    if ispc && isequal(get(hObject,'BackgroundColor'),
    get(0,'defaultUicontrolBackgroundColor'))
        set(hObject,'BackgroundColor','white');
    end
    function edit6_Callback(hObject, eventdata, handles)
    function edit6_CreateFcn(hObject, eventdata, handles)
    if ispc && isequal(get(hObject,'BackgroundColor'),
    get(0,'defaultUicontrolBackgroundColor'))
        set(hObject,'BackgroundColor','white');
    end
    function edit7_Callback(hObject, eventdata, handles)
    function edit7_CreateFcn(hObject, eventdata, handles)
    if ispc && isequal(get(hObject,'BackgroundColor'),
    get(0,'defaultUicontrolBackgroundColor'))
        set(hObject,'BackgroundColor','white');
    end
    function edit8_Callback(hObject, eventdata, handles)
    function edit8_CreateFcn(hObject, eventdata, handles)
    if ispc && isequal(get(hObject,'BackgroundColor'),
    get(0,'defaultUicontrolBackgroundColor'))
        set(hObject,'BackgroundColor','white');
    end
end

```

7.2.2 Detection and Extraction the License Plate Procedure (file: DE_LP.m)

```
function [ candidatePlate ] = DE_LP( originalImage )
imagen = originalImage;
if size(imagen,3)==3
    imagen=rgb2gray(imagen);
end
imagen=imadjust(imagen);
imagen = medfilt2(imagen,[5 5]);
[junk threshold] = edge(imagen,'sobel');
fudgeFactor = .5;
BWs = edge(imagen,'sobel',threshold * fudgeFactor);
BWs = bwareaopen(BWs, 1000);
BWs = imclearborder(BWs,8);
se90 = strel('line', 3, 90);
se0 = strel('line', 3, 0);
BWsdil = imdilate(BWs, [se90 se0]);
BWdfill = imfill(BWsdil, 'holes');
[L Ne]=bwlabel(BWdfill);
maxObject = 0;
for n=1:Ne
    [r,c] = find(L==n);
    if( 0.42<((max(r)-min(r))/(max(c)-min(c))) ...
        && ((max(r)-min(r))/(max(c)-min(c))) < 0.66 ...
        && ((max(r)-min(r))*(max(c)-min(c))*14>16128))
        if(size(originalImage,3)==3)
            canditatedObject =originalImage(min(r):max(r),min(c):max(c),[1,2,3]);
        else
            canditatedObject =originalImage(min(r):max(r),min(c):max(c));
        end
        if((max(r)-min(r))*(max(c)-min(c))>maxObject)
            maxObject = (max(r)-min(r))*(max(c)-min(c));
            candidatePlate = canditatedObject;
        end
    end
end
if(maxObject == 0)
    candidatePlate = 0;
end
end
```

7.2.3 Normalization and Segmentation the License Plate Procedure (file: NS_LP.m)

```
function [ q1Img, q2Img, q3Img, q4Img ] = NS_LP( lpImg )
[lpL,lpW,lpColorScale] = size(lpImg);
x_min = 0;
y_min = 0;
x_width = lpW - 0.158*lpW;
y_width = lpL;
lpBody = imcrop(lpImg,[x_min y_min x_width y_width]);
x_min = 0;
y_min = 0;
x_width = lpW;
y_width = lpL - 0.5*lpL;
lpUp = imcrop(lpBody,[x_min y_min x_width y_width]);
x_min = 0;
y_min = lpL*0.5 + 1;
x_width = lpW;
y_width = lpL - 0.5*lpL;
lpDown = imcrop(lpBody,[x_min y_min x_width y_width]);
[lpUpL,lpUpW,lpColorScale] = size(lpUp);
x_min = 0;
y_min = 0;
x_width = lpUpW - 0.427*lpUpW;
y_width = lpUpL;
q1Img = imcrop(lpUp,[x_min y_min x_width y_width]);
x_min = lpUpW*0.572 + 1;
y_min = 0;
x_width = 0.427*lpUpW;
y_width = lpUpL;
q2Img = imcrop(lpUp,[x_min y_min x_width y_width]);
```

```

[lpDownL,lpDownW,lpColorScale] = size(lpDown);
x_min = 0;
y_min = 0;
x_width = lpDownW - 0.427*lpDownW;
y_width = lpDownL;
q3Img = imcrop(lpDown,[x_min y_min x_width y_width]);
x_min = lpDownW*0.572 + 1;
y_min = 0;
x_width = 0.427*lpDownW;
y_width = lpDownL;
q4Img = imcrop(lpDown,[x_min y_min x_width y_width]);
end

```

7.2.4 Character Segmentation and Recognition Procedure (file: OCR.m)

```

function [ q1Rec, q2Rec, q3Rec, q4Rec] = OCR( q1Img, q2Img, q3Img, q4Img )
AL = importdata('templates_a_letter.mat');
AD = importdata('templates_a_digit.mat');
EL = importdata('templates_e_letter.mat');
ED = importdata('templates_e_digit.mat');
global AL
global AD
global EL
global ED
if size(q1Img,3)==3
    q1Img=rgb2gray(q1Img);
end
q1Img = medfilt2(q1Img,[4 4]);
threshold = graythresh(q1Img);
q1Img = ~im2bw(q1Img,threshold);
q1Img = imclearborder(q1Img,4);
q1Img = bwareaopen(q1Img,50);
seD = strel('diamond',1);
q1Img = imerode(q1Img,seD);
Q1Word = [];
[L Ne] = bwlabel(q1Img);
for n=1:Ne
    [r,c] = find(L==n);
    if(n==1)
        prev_c_min = min(c);
        prev_c_max = max(c);
        prev_r_min = min(r);
        prev_r_max = max(r);
    end
    if( ~(prev_c_min < min(c) && min(c) < prev_c_max ) && (prev_r_min < min(r) && min(r) < prev_r_max ))
        characterImg=q1Img(min(r):max(r),min(c):max(c));
        img_r=imresize(characterImg,[42 24]);
        letter=read_a_digit(img_r);
        Q1Word=[Q1Word ' ' letter];
        prev_c_min = min(c);
        prev_c_max = max(c);
        prev_r_min = min(r);
        prev_r_max = max(r);
    end
end
if size(q2Img,3)==3
    q2Img=rgb2gray(q2Img);
end
q2Img = medfilt2(q2Img,[5 5]);
threshold = graythresh(q2Img);
q2Img = ~im2bw(q2Img,threshold);
q2Img = imclearborder(q2Img,4);
q2Img = bwareaopen(q2Img,50);
seD = strel('diamond',1);
q2Img = imerode(q2Img,seD);
Q2Word = [];
[L Ne] = bwlabel(q2Img);
for n=1:Ne
    [r,c] = find(L==n);
    if(n==1)

```

```

        prev_c_min = min(c);
        prev_c_max = max(c);
        prev_r_min = min(r);
        prev_r_max = max(r);
    end
    if( ~( (prev_c_min < min(c) && min(c) < prev_c_max ) && (prev_r_min < min(r) && min(r)
< prev_r_max )) )
        characterImg=q2Img(min(r):max(r),min(c):max(c));
        img_r=imresize(characterImg,[42 24]);
        letter=read_a_letter(img_r);
        Q2Word=[Q2Word ' ' letter];
        prev_c_min = min(c);
        prev_c_max = max(c);
        prev_r_min = min(r);
        prev_r_max = max(r);
    end
end
if size(q3Img,3)==3
    q3Img=rgb2gray(q3Img);
end
q3Img = medfilt2(q3Img,[5 5]);
threshold = graythresh(q3Img);
q3Img =~im2bw(q3Img,threshold);
q3Img = imclearborder(q3Img,4);
q3Img = bwareaopen(q3Img,50);
seD = strel('diamond',1);
q3Img = imerode(q3Img,seD);
Q3Word = [];
[L Ne] = bwlabel(q3Img);
for n=1:Ne
    [r,c] = find(L==n);
    if(n==1)
        prev_c_min = min(c);
        prev_c_max = max(c);
        prev_r_min = min(r);
        prev_r_max = max(r);
    end
    if( ~( (prev_c_min < min(c) && min(c) < prev_c_max ) && (prev_r_min < min(r) && min(r)
< prev_r_max )) )
        characterImg=q3Img(min(r):max(r),min(c):max(c));
        img_r=imresize(characterImg,[42 24]);
        letter=read_e_digit(img_r);
        Q3Word=[Q3Word ' ' letter];
        prev_c_min = min(c);
        prev_c_max = max(c);
        prev_r_min = min(r);
        prev_r_max = max(r);
    end
end
if size(q4Img,3)==3
    q4Img=rgb2gray(q4Img);
end
q4Img = medfilt2(q4Img,[5 5]);
threshold = graythresh(q4Img);
q4Img =~im2bw(q4Img,threshold);
q4Img = imclearborder(q4Img,4);
q4Img = bwareaopen(q4Img,50);
seD = strel('diamond',1);
q4Img = imerode(q4Img,seD);
Q4Word = [];
[L Ne] = bwlabel(q4Img);
for n=1:Ne
    [r,c] = find(L==n);
    if(n==1)
        prev_c_min = min(c);
        prev_c_max = max(c);
        prev_r_min = min(r);
        prev_r_max = max(r);
    end
    if( ~( (prev_c_min < min(c) && min(c) < prev_c_max ) && (prev_r_min < min(r) && min(r)
< prev_r_max )) )

```

```

        characterImg=q4Img(min(r):max(r),min(c):max(c));
        img_r=imresize(characterImg,[42 24]);
        letter=read_e_letter(img_r);
        Q4Word=[Q4Word ' ' letter];
        prev_c_min = min(c);
        prev_c_max = max(c);
                prev_r_min = min(r);
                prev_r_max = max(r);
    end
end

q1Rec = Q1Word;
q2Rec = Q2Word;
q3Rec = Q3Word;
q4Rec = Q4Word;
end

```